SO-WebServer: User Guide

1. Introduction

This document exists to guide you on how to set up this implementation of a *ConcurrentHTTP Server* system, by compiling it and then executing it.
This user guide will present you with the all compilation options, all execution options, and some examples of the running server.

2. Compilation Instructions

To start, you must compile the source files provided. If you are setting this server up in a Linux system, then the provided Makefile should make this easier; move your current working directory to be in the same directory as the Makefile and run on of the following commands:

- `make` to build the server;
- `make clean && make` to rebuild the server from scratch;
- `make run` to build and run;

Those are the basic commands to compile with Makefile, and basically what you will use most of the time; however there are a lot more commands supported by it, listed below

- `make all` - Build server executable (default)
- `make clean` - Remove object files and executable
- `make run` - Build and start server
- `make test` - Build and run test suite
- `make debug` - Build with debug symbols (-g)
- `make release` - Build with optimizations (-O3)
- `make valgrind` - Build and run under Valgrind
- `make helgrind` - Build and run under Helgrind

In case you are unable to run the Makefile, you can compile the server manually with the following command:

```
gcc -Wall -Wextra -pthread -lrt -o server src/master.c src/worker.c src/http.c src/thread_pool.c src/cache.c src/logger.c src/stats.c src/config.c
```

3. Execution Instructions

To run the server in the most simple way, all you have to do is run the command `./master` in the directory of the executable. Running the server this way will make it so the default values, a.k.a. the ones in server.conf, are the ones used to initialize the server and accept connections. These values are:

Parameter	Default value	Description
PORT	8080	TCP port for HTTP server
TIMEOUT_SECONDS	10	Connection timeout
DOCUMENT_ROOT	../tests	Root directory for serving files
NUM_WORKES	4	Number of worker processes
THREADS_PER_WORKER	10	Thread pool size per worker
MAX_QUEUE_SIZE	100	Connection queue capacity
CACHE_SIZE_MB	10	Maximum cache size per worker
LOG_FILE	access.log	Path to access log
LOG_LEVEL	INFO *	Level of detail of log

* allowed values: DEBUG, INFO, WARN, ERROR. Any other value will be treated as ERROR

Some of these values can be changed in 2 different ways, either overriding them with environment variables or by using one of the command line options when running the server. These command line options have other uses beyond overriding the variables from the configuration file, and they are listed below.

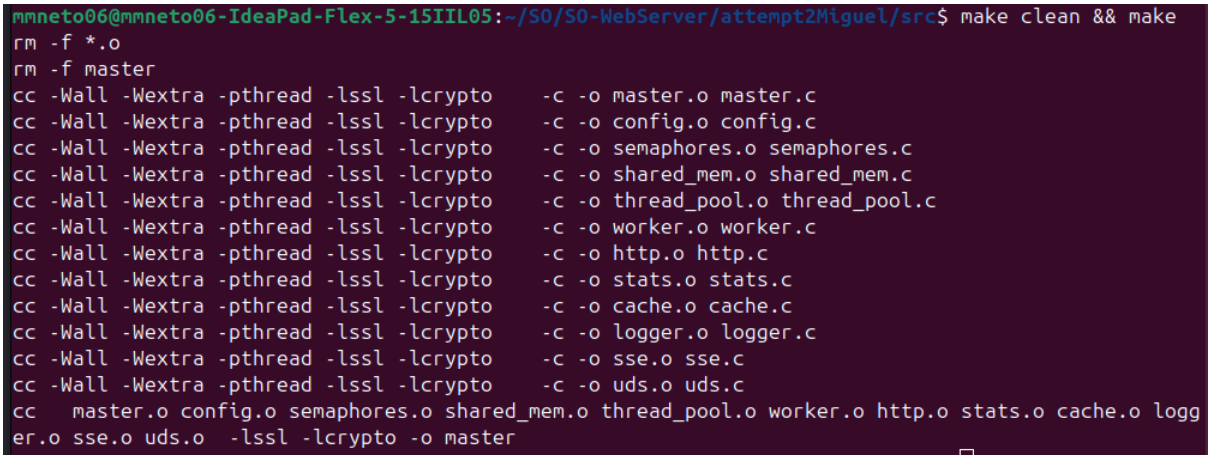
- `-c, --config PATH` : Configuration file path (default: ./server.conf)
- `-p, --port PORT` : Override port to listen on
- `-w, --workers NUM` : Override number of worker processes
- `-t, --threads NUM` : Override number of threads per worker
- `-v, --verbose` : Enable verbose logging
- `-h, --help` : Shows a help message with these options
- `-V, --version` : Show version information

Now you have an up and running server. You can access it with your browser on localhost:PORT (by default PORT=8080, as seen above) or with the curl command on your terminal, and you can stop the server with Ctrl+C or by killing it manually.

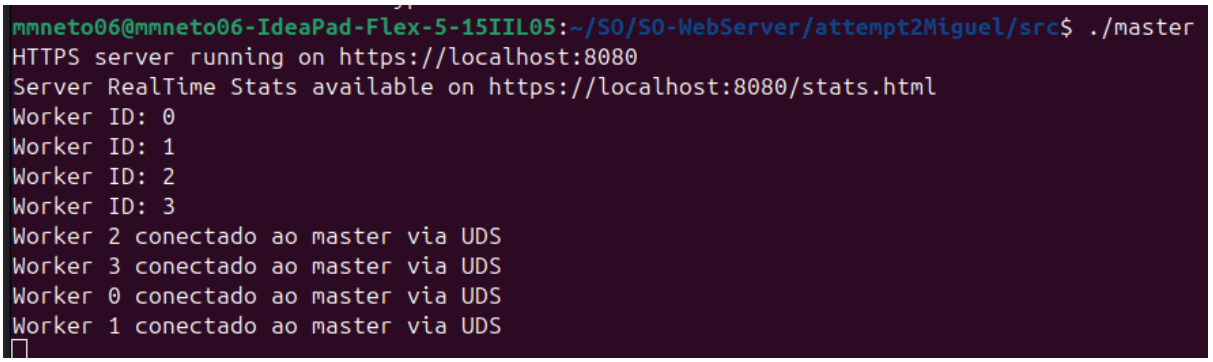
4. Examples

Here are some images that demonstrate compiling and executing the server.

- make clean && make



- running server with default configuration



- running server with 10 workers

```
mmneto06@mmneto06-IdeaPad-Flex-5-15IIL05:~/SO/SO-WebServer/attempt2Miguel/src$ ./master -w 10
HTTPS server running on https://localhost:8080
Server RealTime Stats available on https://localhost:8080/stats.html
Worker ID: 0
Worker ID: 1
Worker ID: 2
Worker ID: 3
Worker ID: 4
Worker ID: 5
Worker ID: 6
Worker ID: 8
Worker ID: 7
Worker 2 conectado ao master via UDS
Worker ID: 9
Worker 1 conectado ao master via UDS
Worker 3 conectado ao master via UDS
Worker 0 conectado ao master via UDS
Worker 5 conectado ao master via UDS
Worker 4 conectado ao master via UDS
Worker 8 conectado ao master via UDS
Worker 7 conectado ao master via UDS
Worker 6 conectado ao master via UDS
Worker 9 conectado ao master via UDS
```

- running server with a different configuration file

```
mmneto06@mmneto06-IdeaPad-Flex-5-15IIL05:~/SO/SO-WebServer/attempt2Miguel/src$ ./master --config /home/mmneto06/Secretária/server.conf
HTTPS server running on https://localhost:3030
Server RealTime Stats available on https://localhost:3030/stats.html
Worker ID: 0
Worker ID: 1
Worker ID: 2
Worker ID: 3
Worker ID: 4
Worker 2 conectado ao master via UDS
Worker 3 conectado ao master via UDS
Worker 0 conectado ao master via UDS
Worker 1 conectado ao master via UDS
Worker 4 conectado ao master via UDS
```